

WCAG 2.2 Rules Engine — Breakdown by Principle

This document contains a development-friendly **rules-engine** broken into four files (as sections below):

- `wcag_perceivable.jsonc`
- `wcag_operable.jsonc`
- `wcag_understandable.jsonc`
- `wcag_robust.jsonc`

Each file is provided in **JSONC** (JSON with comments) for developer readability. Each rule object contains these fields:

- `id` — WCAG Success Criterion identifier (e.g., "1.1.1").
- `name` — short name of the SC.
- `wcag_level` — A / AA / AAA.
- `description` — short description of the SC.
- `automatable` — whether the check is programmatically automatable (true/false).
- `automation_notes` — brief notes on how to automate or limits.
- `selector_checks` — an array of heuristic machine-checks (CSS selectors, DOM checks, or computed-style checks). Each entry can include `selector`, `error`, and `fix`.
- `manual_review_required` — whether human judgement is needed for full compliance.
- `tags` — keywords for grouping and filtering.

```
// wcag_perceivable.jsonc
[
  // 1.1.1 Non-text Content (Level A)
  {
    "id": "1.1.1",
    "name": "Non-text Content",
    "wcag_level": "A",
    "description": "Provide text alternatives for non-text content.",
    "automatable": true,
    "automation_notes": "Detect presence/absence of alt attributes, track <img>, <svg>, <canvas>, <object>, <area>, <input type=\"image\">. Quality of alt text requires human review.",
    "selector_checks": [
      { "selector": "img:not([alt])", "error": "Image missing alt attribute.", "fix": "Add meaningful alt or aria-hidden/role=presentation if decorative." },
      { "selector": "img[alt=\"\"]", "error": "Empty alt attribute (verify decorative).", "fix": "Mark decorative images with role=presentation or aria-hidden if truly decorative." },
      { "selector": "svg[role=\"img\"]:not([aria-label]):not([aria-labelledby])", "error": "SVG with role=img missing accessible name.", "fix":
```

```

"Add aria-label or aria-labelledby or <title> inside SVG." }
],
"manual_review_required": true,
"tags": ["images","alt","non-text"]
},

// 1.2.1 Audio-only and Video-only (Prerecorded) (Level A)
{
  "id": "1.2.1",
  "name": "Audio-only and Video-only (Prerecorded)",
  "wcag_level": "A",
  "description": "Provide text alternatives for prerecorded audio-only and
video-only media.",
  "automatable": true,
  "automation_notes": "Detect <audio> or <video> elements without captions/
transcript; detect mime types likely audio/video. Cannot verify transcript
quality.",
  "selector_checks": [
    { "selector": "audio:not([aria-describedby]):not([data-transcript])",
"error": "Audio element missing transcript or description.", "fix": "Provide a
text transcript or long description." },
    { "selector": "video:not(:has(track[kind=\"captions\"]))", "error": "Video
missing captions.", "fix": "Add <track kind=\"captions\" src=\"...\"> or provide
caption files." }
  ],
  "manual_review_required": true,
  "tags": ["media","captions","transcript"]
},

// 1.2.2 Captions (Prerecorded) (Level A)
{
  "id": "1.2.2",
  "name": "Captions (Prerecorded)",
  "wcag_level": "A",
  "description": "Provide captions for prerecorded audio in synchronized
media.",
  "automatable": true,
  "automation_notes": "Detect <video> elements lacking captions tracks; can
detect presence but not caption quality or synchronicity.",
  "selector_checks": [
    { "selector": "video:not(:has(track[kind=\"captions\"]))", "error":
"Prerecorded video has no captions.", "fix": "Add captions via <track
kind=\"captions\"> or external caption file." }
  ],
  "manual_review_required": true,
  "tags": ["media","captions"]
},

```

```

// 1.2.3 Audio Description or Media Alternative (Prerecorded) (Level A)
{
  "id": "1.2.3",
  "name": "Audio Description or Media Alternative (Prerecorded)",
  "wcag_level": "A",
  "description": "Provide an audio description or alternative for prerecorded
video content where visual information is important.",
  "automatable": false,
  "automation_notes": "Can flag videos that likely convey important visual
info (e.g., no captions only), but deciding need for audio description requires
human judgment.",
  "selector_checks": [],
  "manual_review_required": true,
  "tags": ["media", "audio-description"]
},

// 1.2.4 Captions (Live) (Level AA)
{
  "id": "1.2.4",
  "name": "Captions (Live)",
  "wcag_level": "AA",
  "description": "Provide captions for live audio content in synchronized
media.",
  "automatable": true,
  "automation_notes": "Detect live streams/video elements with aria-live
streams lacking captions. Hard to validate real-time captions are present
programmatically.",
  "selector_checks": [
    { "selector": "video[data-
live=\"true\"]:not(:has(track[kind=\"captions\"]))", "error": "Live video
missing captions.", "fix": "Integrate live captioning or provide live caption
feed." }
  ],
  "manual_review_required": true,
  "tags": ["media", "captions", "live"]
},

// 1.2.5 Audio Description (Prerecorded) (Level AA)
{
  "id": "1.2.5",
  "name": "Audio Description (Prerecorded)",
  "wcag_level": "AA",
  "description": "Provide audio description for prerecorded video content when
necessary.",
  "automatable": false,
  "automation_notes": "Flag likely-needed videos; human judgment required.",
  "selector_checks": [],
  "manual_review_required": true,

```

```

    "tags": ["media","audio-description"]
  },

  // 1.2.6 Sign Language (Prerecorded) (Level AAA)
  {
    "id": "1.2.6",
    "name": "Sign Language (Prerecorded)",
    "wcag_level": "AAA",
    "description": "Provide sign language interpretation for prerecorded audio
content when sign language is the primary means.",
    "automatable": false,
    "automation_notes": "Rarely automatable; flag candidate content.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["media","sign-language"]
  },

  // 1.2.7 Extended Audio Description (Prerecorded) (Level AAA)
  {
    "id": "1.2.7",
    "name": "Extended Audio Description (Prerecorded)",
    "wcag_level": "AAA",
    "description": "Provide extended audio descriptions for prerecorded video
when short descriptions are insufficient.",
    "automatable": false,
    "automation_notes": "Requires content analysis and human judgment.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["media","audio-description"]
  },

  // 1.2.8 Media Alternative (Prerecorded) (Level AAA)
  {
    "id": "1.2.8",
    "name": "Media Alternative (Prerecorded)",
    "wcag_level": "AAA",
    "description": "Provide a media alternative for time-based media.",
    "automatable": false,
    "automation_notes": "Flag presence of media and missing alternatives; human
judgement required.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["media"]
  },

  // 1.2.9 Audio-only (Live) (Level A) -- hypothetical mapping; ensure spec
mapping
  {

```

```

    "id": "1.2.9",
    "name": "Audio-only (Live)",
    "wcag_level": "A",
    "description": "Live audio-only content has a text alternative.",
    "automatable": true,
    "automation_notes": "Detect audio streams and flag missing transcripts/live
captions.",
    "selector_checks": [
      { "selector": "audio[data-live=\"true\"]:not([data-transcript])", "error":
"Live audio missing transcript.", "fix": "Provide live transcript or
captions." }
    ],
    "manual_review_required": true,
    "tags": ["media","live"]
  },

  // 1.3.1 Info and Relationships (Level A)
  {
    "id": "1.3.1",
    "name": "Info and Relationships",
    "wcag_level": "A",
    "description": "Information and relationships conveyed through presentation
can be programmatically determined.",
    "automatable": true,
    "automation_notes": "Detect semantic markup misuse: many <div>/<span> where
headings/lists/tables expected. Flag missing roles/landmarks.",
    "selector_checks": [
      { "selector": "header, nav, main, footer, aside", "error": "Landmarks
presence check (encouraged).", "fix": "Add appropriate HTML5 landmarks or ARIA
roles." },
      { "selector": ":not(:has(h1,h2,h3,h4,h5,h6)):where(section,article)",
"error": "Sectioning element without headings.", "fix": "Add headings to
sectioning elements." }
    ],
    "manual_review_required": true,
    "tags": ["semantics","structure"]
  },

  // 1.3.2 Meaningful Sequence (Level A)
  {
    "id": "1.3.2",
    "name": "Meaningful Sequence",
    "wcag_level": "A",
    "description": "Content must be in a meaningful reading sequence.",
    "automatable": true,
    "automation_notes": "Compare DOM order to visual order heuristics (bounding
boxes) using a headless browser; flag mismatches.",
    "selector_checks": [

```

```

        { "selector": "*", "error": "DOM order differs from visual reading order
        (heuristic).", "fix": "Reorder DOM or use CSS ordering carefully with aria-
        flowto as needed." }
    ],
    "manual_review_required": true,
    "tags": ["reading-order","sequence"]
},

// 1.3.3 Sensory Characteristics (Level A)
{
    "id": "1.3.3",
    "name": "Sensory Characteristics",
    "wcag_level": "A",
    "description": "Instructions should not rely only on sensory characteristics
    of components (shape, size, visual location).",
    "automatable": false,
    "automation_notes": "Flag instructional text containing phrases like 'click
    the red button' or 'select the circle'. Use NLP heuristics to detect. Human
    review needed.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["instructions","content"]
},

// 1.4.1 Use of Color (Level A)
{
    "id": "1.4.1",
    "name": "Use of Color",
    "wcag_level": "A",
    "description": "Color should not be the only way to convey information.",
    "automatable": true,
    "automation_notes": "Detect instances where text is inside elements with
    color-only differences and no text labels or icons; use heuristics and DOM
    analysis.",
    "selector_checks": [
        { "selector": "[style*='color']", "error": "Content uses color to convey
        meaning (heuristic).", "fix": "Provide additional non-color indicators (text
        labels, icons)." }
    ],
    "manual_review_required": true,
    "tags": ["color","visual"]
},

// 1.4.2 Audio Control (Level A)
{
    "id": "1.4.2",
    "name": "Audio Control",
    "wcag_level": "A",

```

```

    "description": "If audio plays automatically for more than 3 seconds,
provide a mechanism to pause/stop/adjust volume.",
    "automatable": true,
    "automation_notes": "Detect autoplay audio/video elements and flag absence
of controls or pause mechanisms.",
    "selector_checks": [
        { "selector": "audio[autoplay]:not([controls]),
video[autoplay]:not([controls])", "error": "Autoplay media without controls.",
"fix": "Add controls or disable autoplay." }
    ],
    "manual_review_required": false,
    "tags": ["media","autoplay"]
},

// 1.4.3 Contrast (Minimum) (Level AA)
{
    "id": "1.4.3",
    "name": "Contrast (Minimum)",
    "wcag_level": "AA",
    "description": "Text and images of text must have a contrast ratio of at
least 4.5:1 (normal) or 3:1 (large text).",
    "automatable": true,
    "automation_notes": "Compute computed styles (color/background-color) via
headless browser (Playwright/Puppeteer) to obtain exact contrast ratios. Inline
style parsing alone is insufficient.",
    "selector_checks": [
        { "selector": "*", "error": "Text with low contrast (ratio < 4.5 for
normal text).", "fix": "Increase contrast or adjust background/text colors." }
    ],
    "manual_review_required": false,
    "tags": ["contrast","color"]
},

// 1.4.4 Resize text (Level AA)
{
    "id": "1.4.4",
    "name": "Resize text",
    "wcag_level": "AA",
    "description": "Text should be resizable up to 200% without loss of content
or functionality.",
    "automatable": false,
    "automation_notes": "Can run automated resize tests in headless browser to
detect clipping/overflow. Full validation often requires visual/human checks
across breakpoints.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["responsive","text"]
},

```

```

// 1.4.5 Images of Text (Level AA)
{
  "id": "1.4.5",
  "name": "Images of Text",
  "wcag_level": "AA",
  "description": "Use real text instead of images of text except for
decoration or necessary cases.",
  "automatable": true,
  "automation_notes": "Detect raster images that look like text (OCR
heuristics) and flag their use. Verify if decorative via alt attribute.",
  "selector_checks": [
    { "selector": "img", "error": "Image used where text preferred (heuristic
via filename or OCR).", "fix": "Replace with actual text where possible." }
  ],
  "manual_review_required": true,
  "tags": ["images","text"]
},

// 1.4.6 Contrast (Enhanced) (Level AAA)
{
  "id": "1.4.6",
  "name": "Contrast (Enhanced)",
  "wcag_level": "AAA",
  "description": "Stricter contrast requirements (7:1 for normal text; 4.5:1
for large).",
  "automatable": true,
  "automation_notes": "Same automation approach as 1.4.3 but different
thresholds.",
  "selector_checks": [],
  "manual_review_required": false,
  "tags": ["contrast","color"]
},

// 1.4.7 Low or No Background Audio (Level A)
{
  "id": "1.4.7",
  "name": "Low or No Background Audio",
  "wcag_level": "A",
  "description": "Background audio should be low volume or have a means to
turn off.",
  "automatable": true,
  "automation_notes": "Detect audio elements flagged as background via aria-
hidden or data attributes; detect autoplay background audio presence.",
  "selector_checks": [
    { "selector": "audio.background, [aria-hidden=\"true\"] audio", "error":
"Background audio may be disruptive.", "fix": "Provide controls or lower volume;
avoid autoplay." }
  ]
}

```



```

    ],
    "manual_review_required": true,
    "tags": ["audio","background"]
  },

  // 1.4.8 Visual Presentation (Level AAA)
  {
    "id": "1.4.8",
    "name": "Visual Presentation",
    "wcag_level": "AAA",
    "description": "Provide control over text spacing, line height,
justification etc.",
    "automatable": false,
    "automation_notes": "Flag absence of user-controllable text settings; heavy
manual review required.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["readability"]
  },

  // 1.4.9 Images of Text (No Exception) (Level AAA)
  {
    "id": "1.4.9",
    "name": "Images of Text (No Exception)",
    "wcag_level": "AAA",
    "description": "Images of text are only used for essential purposes and must
meet criteria.",
    "automatable": true,
    "automation_notes": "Detect images that contain text and flag; manual review
required to validate exception.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["images","text"]
  }
]

```

```

// wcag_operable.jsonc
[
  // 2.1.1 Keyboard (Level A)
  {
    "id": "2.1.1",
    "name": "Keyboard",
    "wcag_level": "A",
    "description": "All functionality is operable through a keyboard
interface.",
    "automatable": true,

```

```

    "automation_notes": "Check that interactive elements are focusable
(tabindex, anchors with href, buttons). Can run automated keyboard navigation
tests with Puppeteer/Playwright.",
    "selector_checks": [
        { "selector": "a[href], button, input, select, textarea, [role=button]",
"error": "Interactive element not focusable via keyboard (heuristic).", "fix":
"Ensure element is natively focusable or add tabindex and keyboard handlers." }
    ],
    "manual_review_required": true,
    "tags": ["keyboard","interaction"]
},

// 2.1.2 No Keyboard Trap (Level A)
{
    "id": "2.1.2",
    "name": "No Keyboard Trap",
    "wcag_level": "A",
    "description": "Users can move focus away from any component using the
keyboard.",
    "automatable": true,
    "automation_notes": "Automated scripts can attempt to tab through UI and
detect if focus becomes trapped; edge-case heavy and sometimes needs manual
testing.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["keyboard","trap"]
},

// 2.1.4 Character Key Shortcuts (Level A)
{
    "id": "2.1.4",
    "name": "Character Key Shortcuts",
    "wcag_level": "A",
    "description": "If single-key shortcuts are provided, they must be
remappable or disabled.",
    "automatable": false,
    "automation_notes": "Detect presence of keydown listeners bound to single
characters via inline scripts heuristics; manual verification needed.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["shortcuts"]
},

// 2.2.1 Timing Adjustable (Level A)
{
    "id": "2.2.1",
    "name": "Timing Adjustable",
    "wcag_level": "A",

```

```

    "description": "Users must be able to extend, adjust, or turn off time
limits.",
    "automatable": false,
    "automation_notes": "Detect timers exposed in JS (heuristic) and presence of
controls; heavy manual testing required.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["timing"]
  },

  // 2.2.2 Pause, Stop, Hide (Level A)
  {
    "id": "2.2.2",
    "name": "Pause, Stop, Hide",
    "wcag_level": "A",
    "description": "Provide user controls to pause, stop or hide moving,
blinking or scrolling content that starts automatically and lasts more than 5
seconds.",
    "automatable": true,
    "automation_notes": "Detect CSS animations, marquee, autoplaying carousels;
check for accessible controls.",
    "selector_checks": [
      { "selector": "[autoplay], [data-carousel], .marquee", "error":
"Autoplaying or moving content without pause/stop control.", "fix": "Provide
accessible controls to pause/stop or disable autoplay by default." }
    ],
    "manual_review_required": true,
    "tags": ["motion", "autoplay"]
  },

  // 2.3.1 Three Flashes or Below Threshold (Level A)
  {
    "id": "2.3.1",
    "name": "Three Flashes or Below Threshold",
    "wcag_level": "A",
    "description": "Content does not contain flashes that could trigger
seizures.",
    "automatable": true,
    "automation_notes": "Analyze animations and video frames for flash frequency
and intensity using heuristic video analysis.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["seizure", "flashing"]
  },

  // 2.4.1 Bypass Blocks (Level A)
  {
    "id": "2.4.1",

```

```

    "name": "Bypass Blocks",
    "wcag_level": "A",
    "description": "Provide a mechanism to skip blocks of content (e.g., skip to
main content).",
    "automatable": true,
    "automation_notes": "Detect skip links or landmark usage (main, nav,
role=main).",
    "selector_checks": [
        { "selector": "a[href^=\"#\"]:contains('skip'), a.skip-link,
[role=\"main\"]", "error": "No skip link or main landmark detected.", "fix":
"Add a visible skip link and landmarks like <main> or role=main." }
    ],
    "manual_review_required": false,
    "tags": ["navigation", "landmarks"]
},

// 2.4.2 Page Title (Level A)
{
    "id": "2.4.2",
    "name": "Page Title",
    "wcag_level": "A",
    "description": "Web pages must have titles that describe topic or purpose.",
    "automatable": true,
    "automation_notes": "Check <title> content and document.title; heuristics
for meaningfulness via length and presence of topic words.",
    "selector_checks": [
        { "selector": "head > title", "error": "Missing or empty <title>.", "fix":
"Add a concise, descriptive <title>." }
    ],
    "manual_review_required": true,
    "tags": ["metadata", "title"]
},

// 2.4.3 Focus Order (Level A)
{
    "id": "2.4.3",
    "name": "Focus Order",
    "wcag_level": "A",
    "description": "Focus order must follow a meaningful sequence that preserves
meaning and operability.",
    "automatable": true,
    "automation_notes": "Simulate keyboard tab order and compare to DOM/visual
order heuristics. Flag anomalies.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["keyboard", "focus"]
},

```

```

// 2.4.4 Link Purpose (In Context) (Level A)
{
  "id": "2.4.4",
  "name": "Link Purpose (In Context)",
  "wcag_level": "A",
  "description": "The purpose of each link can be determined from link text
alone or from the link text together with its programmatically determinable
context.",
  "automatable": true,
  "automation_notes": "Detect vague link texts like 'click here', 'more', or
bare URLs. Provide evidence of context when available.",
  "selector_checks": [
    { "selector": "a:matchesExact('click here'), a:matchesExact('read more'),
a:matches(^http) ", "error": "Non-descriptive link text.", "fix": "Replace link
text with descriptive phrase or add aria-label." }
  ],
  "manual_review_required": true,
  "tags": ["links", "text"]
},

// 2.4.5 Multiple Ways (Level AA)
{
  "id": "2.4.5",
  "name": "Multiple Ways",
  "wcag_level": "AA",
  "description": "Provide more than one way to locate a web page (except where
the page is the result of, or a step in, a process).",
  "automatable": false,
  "automation_notes": "Detect presence of site search, sitemap, and
navigation; manual review for sufficiency.",
  "selector_checks": [],
  "manual_review_required": true,
  "tags": ["navigation"]
},

// 2.4.6 Headings and Labels (Level AA)
{
  "id": "2.4.6",
  "name": "Headings and Labels",
  "wcag_level": "AA",
  "description": "Headings and labels describe the topic or purpose.",
  "automatable": true,
  "automation_notes": "Detect missing headings for sections and labels for UI
controls; heuristic checks for non-descriptive headings like 'Section'.",
  "selector_checks": [
    { "selector": "h1,h2,h3,h4,h5,h6", "error": "Section headings missing or
non-descriptive (heuristic).", "fix": "Use clear descriptive headings for
sections." }
  ]
}

```

```

    ],
    "manual_review_required": true,
    "tags": ["headings", "labels"]
  },

  // 2.4.7 Focus Visible (Level AA)
  {
    "id": "2.4.7",
    "name": "Focus Visible",
    "wcag_level": "AA",
    "description": "Any keyboard operable user interface has a mode of operation
where the keyboard focus indicator is visible.",
    "automatable": true,
    "automation_notes": "Compute focus-visible styles in headless browser;
detect elements with outline: none or no :focus styles.",
    "selector_checks": [
      { "selector": ":focus", "error": "No visible focus indicator for keyboard
focus (heuristic via computed style).", "fix": "Ensure focus styles are visible
(outline, box-shadow, etc.)." }
    ],
    "manual_review_required": true,
    "tags": ["focus", "keyboard"]
  },

  // 2.4.8 Location (Level AA)
  {
    "id": "2.4.8",
    "name": "Location",
    "wcag_level": "AA",
    "description": "Information about the user's location within a set of web
pages is available.",
    "automatable": false,
    "automation_notes": "Detect breadcrumbs or hierarchical headings; manual
review needed to decide sufficiency.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["navigation", "breadcrumbs"]
  },

  // 2.4.9 Link Purpose (Level A) [redundant with 2.4.4 but site-scope]
  {
    "id": "2.4.9",
    "name": "Link Purpose (Link Only)",
    "wcag_level": "A",
    "description": "Link purpose is determinable from the link text alone.",
    "automatable": true,
    "automation_notes": "Similar checks to 2.4.4; include link-only contexts.",
    "selector_checks": [],

```

```

    "manual_review_required": true,
    "tags": ["links"]
  },

  // 2.4.10 Section Headings (Level A)
  {
    "id": "2.4.10",
    "name": "Section Headings",
    "wcag_level": "A",
    "description": "Section headings are used to organize the content.",
    "automatable": true,
    "automation_notes": "Detect long pages lacking h2/h3 headings; flag content
heavy areas with no headings.",
    "selector_checks": [
      { "selector": "article, section", "error": "Sectioning element may lack
heading.", "fix": "Add headings to article/section elements." }
    ],
    "manual_review_required": true,
    "tags": ["headings", "structure"]
  },

  // 2.4.11 Focus Not Obscured (Level A) – WCAG 2.2 new
  {
    "id": "2.4.11",
    "name": "Focus Not Obscured",
    "wcag_level": "A",
    "description": "Focused element should not be obscured by content (e.g.,
overlays) so it is visible to the user.",
    "automatable": true,
    "automation_notes": "Use headless rendering to check if focused element's
bounding box is covered by other elements with higher z-index and non-
transparent backgrounds.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["focus", "visibility"]
  },

  // 2.4.12 Focus Not Obscured (Non-modal) (Level A) – WCAG 2.2 new
  {
    "id": "2.4.12",
    "name": "Focus Not Obscured (Non-modal)",
    "wcag_level": "A",
    "description": "Similar to 2.4.11; ensure focus indication for non-modal UI
is not obscured.",
    "automatable": true,
    "automation_notes": "Same headless browser z-index checks for non-modal
components.",
    "selector_checks": [],

```

```

    "manual_review_required": true,
    "tags": ["focus","visibility"]
  },

  // 2.4.13 Focus Not Obscured (UI Component) (Level A) – WCAG 2.2 new
  {
    "id": "2.4.13",
    "name": "Focus Not Obscured (UI Component)",
    "wcag_level": "A",
    "description": "Ensure focus indicators for UI components are not obscured
by other content.",
    "automatable": true,
    "automation_notes": "Detect overlay stacking and focus visibility for widget
components.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["focus","visibility"]
  },

  // 2.5.1 Pointer Gestures (Level A)
  {
    "id": "2.5.1",
    "name": "Pointer Gestures",
    "wcag_level": "A",
    "description": "Single pointer activation for operations; avoid complex
gestures.",
    "automatable": false,
    "automation_notes": "Detect custom gesture handlers via JS (heuristic);
manual testing with touch required.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["touch","gestures"]
  },

  // 2.5.2 Pointer Cancellation (Level A)
  {
    "id": "2.5.2",
    "name": "Pointer Cancellation",
    "wcag_level": "A",
    "description": "Provide a way to abort a pointer-initiated action before
completion.",
    "automatable": false,
    "automation_notes": "Hard to detect programmatically; flag custom drag/drop
or long-press without cancel mechanism.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["touch"]
  },

```



```

// 2.5.3 Label in Name (Level A)
{
  "id": "2.5.3",
  "name": "Label in Name",
  "wcag_level": "A",
  "description": "Ensure visible labels are included in the accessible name of
UI components.",
  "automatable": true,
  "automation_notes": "Compare visible label text to accessible name computed
from ARIA/label; flag mismatches.",
  "selector_checks": [
    { "selector": "[role], button, input, select, textarea", "error":
"Accessible name differs from visible label (heuristic).", "fix": "Ensure aria-
label/aria-labelledby or <label> matches visible text." }
  ],
  "manual_review_required": true,
  "tags": ["aria","labels"]
},

// 2.5.4 Motion Actuation (Level A)
{
  "id": "2.5.4",
  "name": "Motion Actuation",
  "wcag_level": "A",
  "description": "Functions that rely on motion are also operable by user
interface components or are disabled.",
  "automatable": false,
  "automation_notes": "Flag device-orientation or motion-based handlers;
manual review required.",
  "selector_checks": [],
  "manual_review_required": true,
  "tags": ["motion","device"]
},

// 2.5.7 Dragging Movements (Level A) – WCAG 2.2 new
{
  "id": "2.5.7",
  "name": "Dragging Movements",
  "wcag_level": "A",
  "description": "Functionality that requires dragging must also be operable
by a single pointer or keyboard or provide an alternative.",
  "automatable": false,
  "automation_notes": "Detect drag-and-drop scripts; flag absence of
alternative controls.",
  "selector_checks": [],
  "manual_review_required": true,
  "tags": ["dragdrop","interaction"]
}

```

```

    },

    // 2.5.8 Target Size (Level AA) – WCAG 2.2 new
    {
        "id": "2.5.8",
        "name": "Target Size",
        "wcag_level": "AA",
        "description": "Interactive targets must be at least 24 by 24 CSS pixels or meet exceptions.",
        "automatable": true,
        "automation_notes": "Compute bounding boxes in headless browser; flag elements smaller than threshold unless exceptions apply.",
        "selector_checks": [
            { "selector": "a, button, input, [role=button]", "error": "Interactive target smaller than 24x24 CSS pixels (heuristic).", "fix": "Increase target size or provide alternative spacing." }
        ],
        "manual_review_required": true,
        "tags": ["touch", "target-size"]
    }
]

```

```

// wcag_understandable.jsonc
[
    // 3.1.1 Language of Page (Level A)
    {
        "id": "3.1.1",
        "name": "Language of Page",
        "wcag_level": "A",
        "description": "The default human language of each web page can be programmatically determined.",
        "automatable": true,
        "automation_notes": "Check html[lang] attribute; flag missing or empty lang on root and intra-page lang changes.",
        "selector_checks": [
            { "selector": "html:not([lang])", "error": "Root html element missing lang attribute.", "fix": "Add lang attribute to <html> (e.g., lang=\"en\")." }
        ],
        "manual_review_required": true,
        "tags": ["language"]
    },

    // 3.1.2 Language of Parts (Level AA)
    {
        "id": "3.1.2",
        "name": "Language of Parts",

```

```

    "wcag_level": "AA",
    "description": "Identify any human languages used in content that are
different from the page default.",
    "automatable": true,
    "automation_notes": "Detect lang attributes on elements that differ from
root; heuristics for text language detection can supplement.",
    "selector_checks": [
        { "selector": "[lang]", "error": "Element declares different lang – ensure
correctness.", "fix": "Use correct language codes on elements containing
different-language content." }
    ],
    "manual_review_required": true,
    "tags": ["language"]
},

// 3.2.1 On Focus (Level A)
{
    "id": "3.2.1",
    "name": "On Focus",
    "wcag_level": "A",
    "description": "When any user interface component receives focus, it does
not initiate a change of context.",
    "automatable": true,
    "automation_notes": "Simulate focus events and observe side-effects that
change context (navigation, modals). Heuristic-based; manual review required for
business logic.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["focus", "behavior"]
},

// 3.2.2 On Input (Level A)
{
    "id": "3.2.2",
    "name": "On Input",
    "wcag_level": "A",
    "description": "Changing the setting of any user interface component does
not automatically cause a change of context unless the user has been advised.",
    "automatable": true,
    "automation_notes": "Detect input handlers that trigger navigation or major
changes without warning; use headless interaction scripts.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["forms", "behavior"]
},

// 3.2.3 Consistent Navigation (Level AA)
{

```

```

    "id": "3.2.3",
    "name": "Consistent Navigation",
    "wcag_level": "AA",
    "description": "Navigational mechanisms that are repeated on multiple pages
must be consistent.",
    "automatable": false,
    "automation_notes": "Site-sweep necessary: compare navigation structures
across pages to detect inconsistency; manual review required to decide
acceptability.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["navigation","consistency"]
  },

  // 3.2.4 Consistent Identification (Level AA)
  {
    "id": "3.2.4",
    "name": "Consistent Identification",
    "wcag_level": "AA",
    "description": "Components with the same functionality should be identified
consistently across the site.",
    "automatable": false,
    "automation_notes": "Aggregate component roles and labels across site pages
and flag inconsistencies; may need manual validation.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["consistency"]
  },

  // 3.2.6 Consistent Help (Level A) – WCAG 2.2 new
  {
    "id": "3.2.6",
    "name": "Consistent Help",
    "wcag_level": "A",
    "description": "Help mechanisms should be consistent and available where
needed.",
    "automatable": false,
    "automation_notes": "Detect presence of help links or help UI in consistent
locations; manual review required to ensure sufficiency.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["help","usability"]
  },

  // 3.3.1 Error Identification (Level A)
  {
    "id": "3.3.1",
    "name": "Error Identification",

```

```

    "wcag_level": "A",
    "description": "When an input error is detected, the error is identified and
described to the user.",
    "automatable": true,
    "automation_notes": "Inspect form validation patterns and aria-invalid
usage; attempt submitting forms programmatically to observe server/client errors
and messages.",
    "selector_checks": [
        { "selector": "input[aria-invalid=\"true\"]", "error": "Form control
indicates error but no visible description found (heuristic).", "fix": "Provide
descriptive error messages with aria-describedby." }
    ],
    "manual_review_required": true,
    "tags": ["forms", "validation"]
},

// 3.3.2 Labels or Instructions (Level A)
{
    "id": "3.3.2",
    "name": "Labels or Instructions",
    "wcag_level": "A",
    "description": "Labels or instructions are provided when content requires
user input.",
    "automatable": true,
    "automation_notes": "Check input elements for associated <label>, aria-
label, or aria-labelledby. Use heuristics for placeholders as insufficient.",
    "selector_checks": [
        { "selector": "input:not([type=hidden]):not([aria-label]):not([aria-
labelledby]):not([id])", "error": "Form control missing accessible label.",
"fix": "Add <label for=...> or aria-label/aria-labelledby." }
    ],
    "manual_review_required": true,
    "tags": ["forms", "labels"]
},

// 3.3.3 Error Suggestion (Level AA)
{
    "id": "3.3.3",
    "name": "Error Suggestion",
    "wcag_level": "AA",
    "description": "If an input error is detected, suggestions for correction
are provided.",
    "automatable": false,
    "automation_notes": "Attempt to detect existence of corrective suggestions
in error messages; manual review required for quality.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["forms", "validation"]
}

```

```

    },

    // 3.3.4 Error Prevention (Legal, Financial, Data) (Level AA)
    {
        "id": "3.3.4",
        "name": "Error Prevention (Legal, Financial, Data)",
        "wcag_level": "AA",
        "description": "For transactions, ensure errors can be reviewed, corrected,
and confirmed before final submission.",
        "automatable": false,
        "automation_notes": "Simulate form flows to detect review/confirmation
steps; manual review needed.",
        "selector_checks": [],
        "manual_review_required": true,
        "tags": ["forms", "transactions"]
    },

    // 3.3.7 Redundant Entry (Level A) – WCAG 2.2 new
    {
        "id": "3.3.7",
        "name": "Redundant Entry",
        "wcag_level": "A",
        "description": "Reduce repeated or redundant data entry across steps when
possible.",
        "automatable": false,
        "automation_notes": "Detect repeated fields across multi-step forms;
heuristic flagging and manual review.",
        "selector_checks": [],
        "manual_review_required": true,
        "tags": ["forms", "usability"]
    },

    // 3.3.8 Accessible Authentication (Level A) – WCAG 2.2 new
    {
        "id": "3.3.8",
        "name": "Accessible Authentication",
        "wcag_level": "A",
        "description": "Authentication mechanisms must provide accessible
alternatives to cognitive or sensory challenges (e.g., CAPTCHA alternatives).",
        "automatable": false,
        "automation_notes": "Detect presence of CAPTCHAs and flag absence of
accessible alternatives like audio/text alternatives or fallback methods.",
        "selector_checks": [
            { "selector": "iframe[src*='captcha'], .g-recaptcha", "error": "CAPTCHA
detected; ensure accessible alternative is available.", "fix": "Provide
accessible authentication alternatives (SMS, email, backup codes)."}
        ],
        "manual_review_required": true,
    }

```

```

    "tags": ["authentication","captcha"]
  },

  // 3.3.9 Accessible Authentication (Convincing non-bypass) (Level A) – WCAG
  2.2 new
  {
    "id": "3.3.9",
    "name": "Accessible Authentication (Convincing non-bypass)",
    "wcag_level": "A",
    "description": "Authentication should not rely on user memory; provide
alternatives and assistance.",
    "automatable": false,
    "automation_notes": "Flag patterns like forced complex password without
assistance; manual review needed.",
    "selector_checks": [],
    "manual_review_required": true,
    "tags": ["authentication"]
  }
]

```

```

// wcag_robust.jsonc
[
  // 4.1.1 Parsing (Level A)
  {
    "id": "4.1.1",
    "name": "Parsing",
    "wcag_level": "A",
    "description": "Ensure content can be parsed by user agents; documents must
not contain major syntax errors that prevent assistive technologies from
determining relationships between elements.",
    "automatable": true,
    "automation_notes": "Run HTML validators (W3C) and ARIA validator
heuristics; flag unclosed tags, duplicate IDs, invalid ARIA attributes.",
    "selector_checks": [
      { "selector": "[role][aria-*]", "error": "Suspicious ARIA usage detected
(heuristic).", "fix": "Validate ARIA roles/attributes against element semantics
and remove invalid ones." },
      { "selector": "[id]", "error": "Duplicate IDs detected (validator).",
"fix": "Ensure unique id values across document." }
    ],
    "manual_review_required": false,
    "tags": ["validation","aria"]
  },

  // 4.1.2 Name, Role, Value (Level A)
  {

```

```

    "id": "4.1.2",
    "name": "Name, Role, Value",
    "wcag_level": "A",
    "description": "For UI components, the name and role must be
programmatically determinable; states and properties should be settable
programmatically.",
    "automatable": true,
    "automation_notes": "Compute accessible name using platform algorithm (a11y-
name) and compare with visible label; check role correctness.",
    "selector_checks": [
      { "selector": "[role]:not([aria-label]):not([aria-labelledby])", "error":
"Element with role missing accessible name.", "fix": "Provide aria-label or
aria-labelledby or visible label." }
    ],
    "manual_review_required": true,
    "tags": ["aria", "name", "role"]
  },

  // 4.1.3 Status Messages (Level AA)
  {
    "id": "4.1.3",
    "name": "Status Messages",
    "wcag_level": "AA",
    "description": "Changes in content should be programmatically determinable
through role=\"status\" or aria-live regions.",
    "automatable": true,
    "automation_notes": "Detect dynamic updates and presence of appropriate ARIA
live regions (aria-live=polite/assertive) or role=status.",
    "selector_checks": [
      { "selector": "[aria-live], [role=\"status\"]", "error": "No aria-live or
status region found for dynamic updates (heuristic).", "fix": "Add aria-live or
status regions for important dynamic updates." }
    ],
    "manual_review_required": true,
    "tags": ["dynamic", "aria"]
  }
]

```

Notes & Next Steps

- These JSONC files are intentionally developer-oriented: they provide both machine-checkable selectors and guidance about automation limits.
- For robust automation (contrast, focus-obscured, target sizes, visual order) you should integrate a **headless browser** (Playwright/Puppeteer) to compute layout, bounding boxes, and computed styles.

- For site-level aggregation, run each page against these rule files and store results; compute site-wide metrics per `rule_id`.
- I can now:
- Export these blocks into separate `.jsonc` files in a zip for you, or
- Convert JSONC into pure JSON (removing comments) for production engines, or
- Generate a Python/Node skeleton that consumes these rules and runs checks against live pages (including Playwright integration for computed checks).

Tell me which follow-up you want (export, convert to JSON, runnable skeleton, or site aggregator script).